DeCODER distribution file-type assay

Version 2 | no LIMIT | 8 October 2026

Version 2 repeats the assay with the LIMIT clause removed. Each catalog was queried on its own endpoint and the results were not added together. A distribution is a schema:DataDownload. The file-type estimate uses schema:contentUrl only, not schema:url and not schema:encodingFormat. The unbounded SELECT returned the same distinct URL counts as a separate COUNT(DISTINCT ?url), so the endpoint did not truncate this result set.

Query

```
PREFIX schema: <https://schema.org/>
SELECT DISTINCT ?url WHERE {
  GRAPH ?g {
    ?dist a schema:DataDownload ;
          schema:contentUrl ?url .
  }
}
```

No LIMIT. Object counts used the same pattern with COUNT(DISTINCT ?dist) and COUNT(DISTINCT ?url), also with no LIMIT.

How a URL was typed

The path is taken before the query string and percent-decoded. If the last path segment ends in a known file suffix, that suffix is the type. Otherwise a query-parameter value is used when that value is itself a filename (Nextcloud links such as files=tile.tif). Compound names such as .csv.gz stay compound. Service scripts (.php, .do, .exe) are not treated as file types. Anything else is unknown. This is an estimate from the URL string. The bytes were not downloaded.

What was in the graphs

Catalog	Distinct contentUrl	DataDownload objects
deepoceans	33,744	154,333
ecoforecast	1,122	2,022
earthsurface	5,211	5,331
facetsearch	50,274	147,554

Endpoints: deepoceans, ecoforecast, and earthsurface on https://qlever.geocodes-aws.earthcube.org/graphspace/<name> ; facetsearch on https://qlever.geocodes-aws-dev.earthcube.org/graphspace/facetsearch. Object counts are higher than URL counts because many downloads share a URL. These totals match version 1; the earlier LIMIT 200000 had not cut any of these sets.

deepoceans

https://qlever.geocodes-aws.earthcube.org/graphspace/deepoceans

Type from URL text	Distinct URLs	Share
unknown	14,254	42.2%
.csv	8,456	25.1%
.zip	5,905	17.5%
.nc	3,648	10.8%
.txt	1,054	3.1%
other recognized suffix	427	1.3%

Other suffixes are mostly .html, .gz, .pdf, .xlsx, .rar, .xml, and .tgz. No URL contains parquet. Unknown includes 7,579 URLs on a bco-dmo.org host, 5,531 IPT archive.do links, and 462 WMS/OWS endpoints.

ecoforecast

https://qlever.geocodes-aws.earthcube.org/graphspace/ecoforecast

Type from URL text	Distinct URLs	Share
unknown	695	61.9%
.json	403	35.9%
.r	17	1.5%
.rmd	5	0.4%
.csv	2	0.2%

No contentUrl ends in .parquet. Of the unknown URLs, 354 are GitHub or STAC browser pages, 220 are S3 paths with no file suffix, and 117 are S3 paths whose text contains bundled-parquet but still has no .parquet suffix. A separate no-LIMIT count of encodingFormat application/x-parquet is 337 distinct contentUrls, so the URL text underestimates parquet.

earthsurface

https://qlever.geocodes-aws.earthcube.org/graphspace/earthsurface

Type from URL text	Distinct URLs	Share
.tif	4,609	88.4%
.gpkg	460	8.8%
unknown	123	2.4%
other recognized suffix	19	0.4%

Almost all .tif (4,601 of 4,609) and the .gpkg names sit in the query string (files=...), not the path. The other suffixes are .txt (6), .html (5), .zip (5), .nc, .pdf, and .xml. Unknown is share links, directories, and DOI landing pages. No parquet.

facetsearch

https://qlever.geocodes-aws-dev.earthcube.org/graphspace/facetsearch

Type from URL text	Distinct URLs	Share
unknown	34,113	67.9%
.zip	5,704	11.3%
.txt	5,497	10.9%
.nc	3,571	7.1%
.csv	1,236	2.5%
other recognized suffix	153	0.3%

Unknown is mostly IODP secureGetObj.php?id=... (26,155) and IPT archive.do links (5,437), plus other .php links (1,874), WMS/OWS (285), and DOI pages (159). No parquet. This graph overlaps some ocean URLs; it was not joined with deepoceans.

Reading the unknown bucket

unknown means the URL text has no recognized filename suffix. It is not a claim that the bytes have no format. IPT archive.do links are download endpoints whose URLs do not say .zip. IODP secureGetObj.php links are opaque ids. S3 forecast paths often omit .parquet even when encodingFormat says parquet. A directory URL or a DOI is a landing place, not a file suffix.